

A/B Testing

Compare two variants with a live audience

PROFESSIONAL DEFINITION

A/B testing is a controlled online experiment that randomly shows variant A or B to comparable groups of real users and measures which performs better on a defined metric. Randomisation lets teams attribute the difference to the change itself, not to opinion or chance.

WHO DEVELOPED IT

Rooted in Fisher's randomised experiments; scaled to the web by companies like Google, Amazon and Microsoft, and codified by Ron Kohavi and colleagues.

WHY IT WAS DEVELOPED

Opinions about which design is better are unreliable. Controlled experiments provide causal evidence, so decisions rest on measured user behaviour.

HOW IT IS USED

Define a hypothesis and metric, split traffic randomly between variants, run until statistically significant, and ship the winner — guarding against common pitfalls.

WHEN IT IS USED

The Test phase and ongoing optimisation of live products.

BY WHOM IT IS USED

Product managers, data scientists and growth teams.

INNOVATION RELATIVE TO PRIOR PRACTICE

It brings **causal, statistical rigour** to design decisions at scale, turning product development into continuous experimentation.

RELATED FRAMEWORKS IN THIS COURSE

Minimum Viable Product (MVP) *Slibrary 8*

Usability Testing *Slibrary 9*

Define Success *Slibrary 10*

SOURCES (HARVARD · CITE THEM RIGHT)

Kohavi, R., Tang, D. and Xu, Y. (2020) *Trustworthy Online Controlled Experiments*. Cambridge: Cambridge University Press.

Thomke, S.H. (2020) *Experimentation Works*. Boston: Harvard Business Review Press.

FIGURE 1 · RANDOMISED CONTROLLED SPLIT

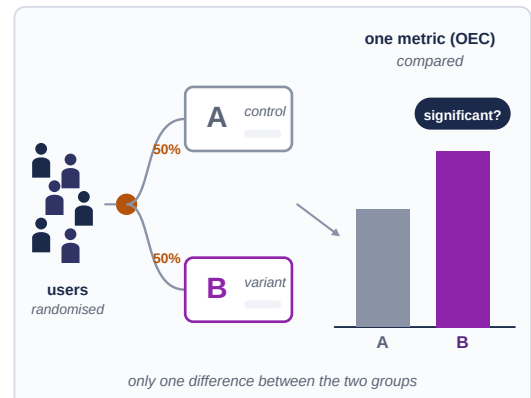


Figure 1. Users are split at random between control and variant; the difference in one metric is tested for significance.